MTH2003 Feedback Sheet 1

These questions are designed for "formative assessment". They are to check your understanding. You should be able to do them on your own, after having done the examples sheet and the tutorial sheet. Your solutions to these questions should be submitted as a pdf via ELE before 12:00 on Friday of week 3 . These questions will be marked by the tutor of your group tutorial, and feedback given to you.

1. For the differential equation
$$t\frac{\mathrm{d}y}{\mathrm{d}t} + y + 4t = 0\,,$$
determine which of the following three methods of solution are applicable:

   [i] the method of separation of variables,
   [ii] the method for a homogeneous first-order ODE,
   [iii] using an integrating factor for a linear first-order ODE.

   Apply all methods that are applicable and derive the solution in each case. Identify the intervals on which your solutions are defined. If a method is not applicable, briefly state why.

2. Consider the ODE
$$\frac{dy}{dt} = -\frac{4}{y}\,,$$
with the initial condition (A) $y(1) = 1$ and (B) $y(-1) = -1$.

   [i] Determine for what pairs (t_0, y_0) the initial value problem $y(t_0) = y_0$ is guaranteed to have locally
   - a solution by Peano's theorem
   - a unique solution by Picard's theorem.

   [ii] Use the method of separation of variables to find any special solutions and typical solutions. For each solution, identify the maximal interval in which it is defined, and the maximal interval in which it is unique.
   [iii] Sketch a graph showing a representative set of typical solutions and any special solutions. Use this diagram to identify possible points of uniqueness violation. Compare these results with your prediction in part [i].
   [iv] Find solutions satisfying the specified initial conditions, and determine the maximal intervals where they exist, and maximal intervals where they are unique. Sketch the graphs of these solutions, indicating the initial points.